

Digital System Assignment

SIMULATE FLASH (3 BIT) AND INTEGRATING TYPE ADCs (8 BIT)

EE25BTECH11011 – B. Navya

1 Introduction

Analog to Digital Converters (ADCs) are used to convert analog signals into digital form. They are widely used in electronics systems.

In this experiment, two types of ADCs are studied:

- **3-bit Flash ADC**
- **8-bit Integrating (Dual-Slope) ADC**

The circuits are designed and simulated using ngspice. The outputs are checked for different input voltages and compared.

PART A: 3-BIT FLASH ADC

1.1 Operating Principle

A Flash ADC converts the input voltage into digital output very fast. It uses multiple comparators working in parallel.

For a 3-bit ADC:

- Number of levels = $2^3 = 8$
- Number of comparators = $2^3 - 1 = 7$

A resistor ladder is used to divide the reference voltage into equal steps. Each comparator compares the input voltage with a reference level and produces output. These outputs are then converted into binary using an encoder.

1.2 Resistor Ladder Node Voltages

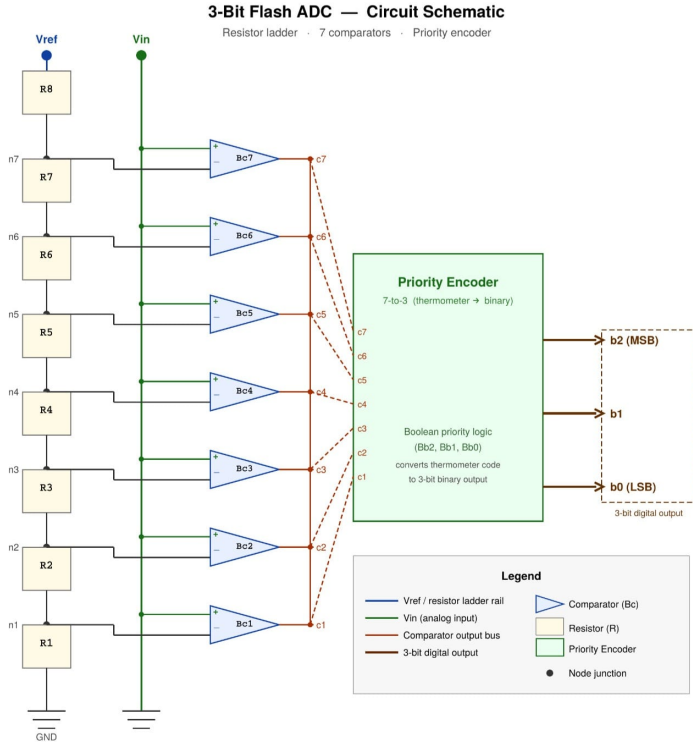
The resistor ladder divides the reference voltage into equal steps.

For $V_{ref} = 5V$ and 8 resistors, the voltage step size is:

$$\Delta V = \frac{V_{ref}}{2^N} = \frac{5}{8} = 0.625V \quad (1)$$

The node voltages are shown below:

Node	Voltage (V)	Comparator
n_1	0.625	c_1
n_2	1.250	c_2
n_3	1.875	c_3
n_4	2.500	c_4
n_5	3.125	c_5
n_6	3.750	c_6
n_7	4.375	c_7



1.3 Circuit Architecture

The three functional blocks are:

1. **Resistor Ladder:** Eight $1\text{ k}\Omega$ resistors in series between V_{ref} and GND generate the seven threshold voltages.
2. **Comparator Bank:** Seven ideal behavioral voltage sources using the unit step $u(\cdot)$ act as single-bit comparators, producing a *thermometer code*.
3. **Priority Encoder:** Boolean logic converts the 7-bit thermometer code $(c_1, \dots, c_7)$ to the 3-bit natural binary output (b_2, b_1, b_0) .

1.4 Encoder Boolean Equations

The encoder converts comparator outputs into binary output bits.

The Boolean expressions used are:

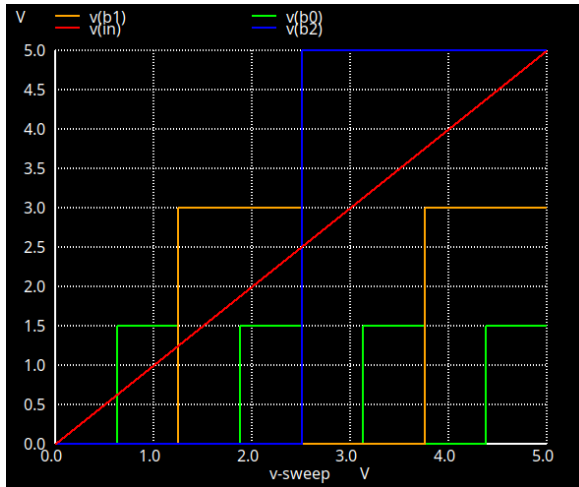
$$b_2 = c_4 \quad (2)$$

$$b_1 = c_6 + c_2 \overline{c_4} \quad (3)$$

$$b_0 = c_7 + c_5 \overline{c_6} + c_3 \overline{c_4} + c_1 \overline{c_2} \quad (4)$$

These equations are implemented in ngspice using behavioral sources. Different voltage levels are used for each bit to clearly observe them in the plot.

1.5 Simulation Waveform



1.6 NGspice Simulation Code

```
1 3-BIT FLASH ADC PROJECT
2 Vin in 0 DC 0
3 Vref vref 0 DC 5
4
5 R8 vref n7 1k
6 R7 n7 n6 1k
7 R6 n6 n5 1k
8 R5 n5 n4 1k
9 R4 n4 n3 1k
10 R3 n3 n2 1k
11 R2 n2 n1 1k
12 R1 n1 0 1k
13
14 Bc1 c1 0 V = 1*u(V(in) - V(n1))
15 Bc2 c2 0 V = 1*u(V(in) - V(n2))
16 Bc3 c3 0 V = 1*u(V(in) - V(n3))
17 Bc4 c4 0 V = 1*u(V(in) - V(n4))
18 Bc5 c5 0 V = 1*u(V(in) - V(n5))
19 Bc6 c6 0 V = 1*u(V(in) - V(n6))
20 Bc7 c7 0 V = 1*u(V(in) - V(n7))
21
22 Bb2 b2 0 V = 5 * u(V(c4) - 0.5)
23 Bb1 b1 0 V = 3 * u((V(c6) + V(c2)*(1-V(c4))) - 0.1)
24 Bb0 b0 0 V = 1.5 * u((V(c7) + V(c5)*(1-V(c6)) + V(c3)*(1-V(c4))
    + V(c1)*(1-V(c2))) - 0.1)
25
26 .dc Vin 0 5 0.01
27
28 .control
29     run
30     plot v(in) v(b2) v(b1) v(b0)
31 .endc
32 .end
```

Waveform Interpretation

The plot contains four traces on the same axes, deliberately scaled to different amplitudes for clarity:

- **Red — $v(in)$:** The analog input ramp increases linearly from 0 V to 5 V, confirming correct stimulus.
- **Blue — $v(b_2)$, MSB (amplitude 5 V):** Remains low until V_{in} crosses the midpoint 2.5 V, then switches high and stays high for the upper half of the

input range. This is the expected behaviour of the MSB in a natural-binary Flash ADC.

- **Orange — $v(b_1)$, middle bit (amplitude 3 V):** Rises to 3 V at $V_{in} = 1.25$ V, falls back at $V_{in} = 2.5$ V, rises again at $V_{in} = 3.75$ V, and remains high to 5 V. Two transitions in the lower half and one additional transition in the upper half are consistent with a Gray-to-binary decoded b_1 .
- **Green — $v(b_0)$, LSB (amplitude 1.5 V):** Toggles at every $\Delta V = 0.625$ V step, producing four complete high pulses across the full input range. This highest toggle rate is characteristic of the LSB.

All transitions are sharp (ideal comparators), confirming that the behavioral $u(\cdot)$ model is functioning correctly with no metastability or glitch artifacts.

1.7 Input vs. Digital Output Truth Table

The following table is read directly from the waveform by evaluating each bit at the centre of each quantisation interval.

Input range (V)	b_2	b_1	b_0	Binary	Decimal
$0.000 \leq V_{in} < 0.625$	0	0	0	000	0
$0.625 \leq V_{in} < 1.250$	0	0	1	001	1
$1.250 \leq V_{in} < 1.875$	0	1	0	010	2
$1.875 \leq V_{in} < 2.500$	0	1	1	011	3
$2.500 \leq V_{in} < 3.125$	1	0	0	100	4
$3.125 \leq V_{in} < 3.750$	1	0	1	101	5
$3.750 \leq V_{in} < 4.375$	1	1	0	110	6
$4.375 \leq V_{in} \leq 5.000$	1	1	1	111	7

PART B: 8-BIT INTEGRATING (DUAL-SLOPE) ADC

Operating Principle

The Dual-Slope ADC converts V_{in} by integrating it for a fixed time and then de-integrating with $-V_{ref}$ until the integrator returns to zero. The counter value at that instant encodes the digital output.

Phase 1 — Integration (fixed $N_{max} = 256$ clock cycles):

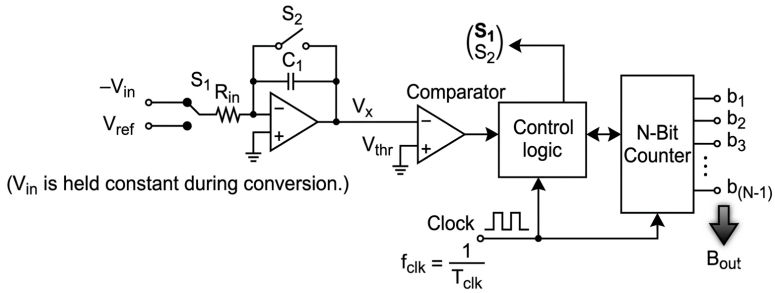
$$V_{int}(T_1) = \frac{V_{in} T_1}{RC} \quad (5)$$

Phase 2 — De-integration (variable N cycles):

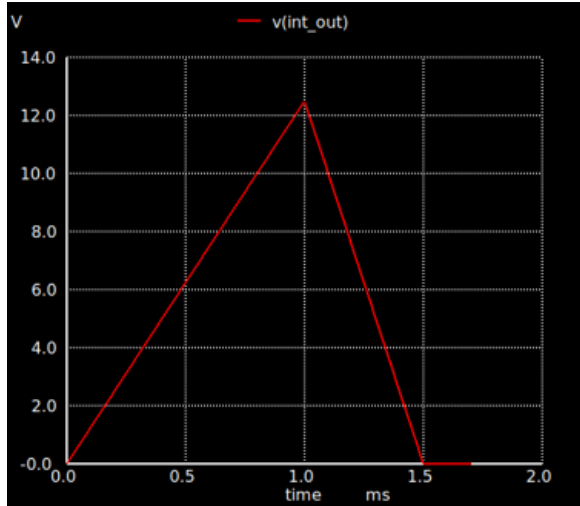
$$\frac{V_{in} T_1}{RC} = \frac{V_{ref} T_2}{RC} \Rightarrow N = \frac{V_{in}}{V_{ref}} \times N_{max} \quad (6)$$

$$D = \frac{V_{in}}{V_{ref}} \times 2^8 = \frac{V_{in}}{12} \times 256 \quad (7)$$

Circuit:



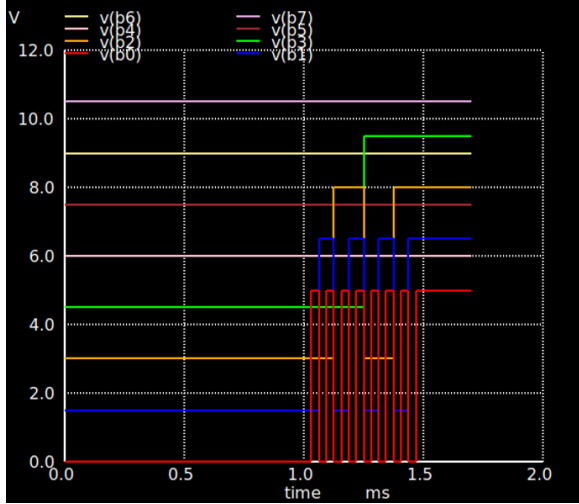
Simulation Results:



- The integrator output rises linearly from 0 V to approximately +12.5 V over the fixed integration interval $T_1 = 1$ ms. This confirms a constant V_{in} being integrated.
- During de-integration (T_1 to $T_2 \approx 1.5$ ms), the output falls linearly back to 0 V under the fixed reference V_{ref} .
- The slope of the rising ramp is $\frac{V_{in}}{RC}$ and the falling ramp is steeper as $V_{ref} > V_{in}$ (de-integration is faster).
- The total de-integration time $T_2 - T_1 = 0.5$ ms, compared to $T_1 = 1$ ms, gives:

$$\frac{V_{in}}{V_{ref}} = \frac{T_2}{T_1} = \frac{0.5}{1} = 0.5 \implies V_{in} = 0.5 \times V_{ref}$$

Digital Output Waveform:



- **b0 (LSB, red):** Toggles at the highest frequency â every clock cycle. This is expected for the least significant bit.
- **b1 (blue):** Toggles at half the frequency of b0.
- **b2, b3 (green):** Toggle at progressively lower frequencies.
- **b4, b5, b6, b7 (MSBs):** Remain mostly high or low, changing only once or a few times during the counting window â characteristic of MSBs.
- The counting begins at approximately $t = 1$ ms and stops at $t \approx 1.5$ ms when the integrator output reaches zero, as controlled by the comparator.

Calculation of Digital Output

From the simulation:

$$T_1 = 1 \text{ ms} \quad (\text{fixed integration time, } N_{\max} = 256 \text{ counts})$$

$$T_2 = 0.5 \text{ ms} \quad (\text{measured de-integration time})$$

$$N = \frac{T_2}{T_1} \times N_{\max} = \frac{0.5}{1} \times 256 = \mathbf{128} = \mathbf{100000000_2}$$

Input Voltage vs. Digital Output ($V_{ref} = 12\text{ V}$, 8-bit)

Vin (V)	Decimal	Binary Output
0.0	0	00000000
1.5	32	00100000
3.0	64	01000000
4.5	96	01100000
6.0	128	10000000
7.5	160	10100000
9.0	192	11000000
10.5	224	11100000
12.0	255	11111111

Comparison: Flash ADC vs Integrating ADC

Feature	Flash ADC	Integrating ADC
Speed	Very fast	Slow
Accuracy	Medium	High
Hardware	Complex (many comparators)	Simple